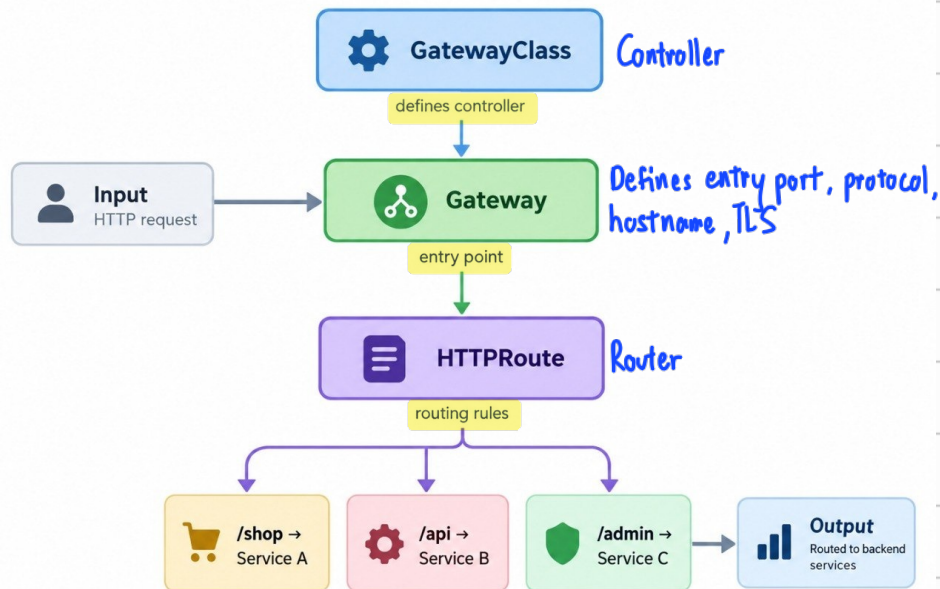


# Gateway API

Gateway API is a family of API resources. It helps to exposing application to external network safely. \* Successor of Ingress.

## Kubernetes Gateway API Flow



## Why?

- Multi-team environment

Platform team: Gateway    Dev team: HTTP routes    Infra team: Gateway Controller

- Introduction of controller [Gateway Class]

Portability ↑ [more versatile], no more rewrite when changing Nginx, Traefik, GKE etc

- Support advance features

Weight attribute in **httpRoute** ⇒ Canary deployment { 90% Old version, 10% New version }

## Request Flow from user to service

1. User open **www.example.com**
2. CloudFlare store A record, match DNS
3. Request reach gateway. ← Gateway class defined the controller to use [Nginx / traefik / gateway]
4. Gateway check **HTTP route**
5. Gateway forwards the request to the correct service.

## Gateway API setup

1. Install a gateway controller [It comes along with Gateway CustomResourceDefinition [CRD]]
  - By go to kubernetes documentation → Gateway Controller's own documentation
2. Apply the gateway controller
3. Create Gateway
4. Create HTTPRoute
5. Add DNS record in Public DNS Provider [CloudFlare]

## HTTPRoute Example

```
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: bullet-train-route
  namespace: jp-bullet-train-gtw
spec:
  parentRefs:
    - name: bullet-train-gateway
  hostnames:
    - bullet.train.io
  rules:
    - matches:
        - path:
            type: PathPrefix
            value: /available
      backendRefs:
        - name: available
          namespace: jp-bullet-train-app-prod
          port: 80
    - matches:
        - path:
            type: PathPrefix
            value: /books
      backendRefs:
        - name: books
          namespace: jp-bullet-train-app-prod
          port: 80
    - matches:
        - path:
            type: PathPrefix
            value: /travellers
      backendRefs:
        - name: travellers
          namespace: jp-bullet-train-app-prod
          port: 80
```

Reference to parent gateway

Match every URL start with this prefix  
Eg: /available/train

Sent the traffic to service named book

### 3 types of path

1. Path Prefix
2. Exact
3. Regular Expression

Can add weight: 90 or 10 for canary deployment. # 90/10% of traffic

## Gateway Example

```
apiVersion: gateway.networking.k8s.io/v1
kind: Gateway
metadata:
  name: bullet-train-gateway
  namespace: jp-bullet-train-gtw
spec:
  gatewayClassName: nginx
  listeners:
    - name: https
      protocol: HTTPS
      port: 443
      hostname: bullet.train.io
      tls:
        mode: Terminate
        certificateRefs:
          - kind: Secret
            group: ""
            name: bullet-train-tls
```

decrypt HTTPS by TLS in gateway  
using defined TLS secret

## ReferenceGrant [Secure Practise]

A kind, to allow cross-namespaces references.

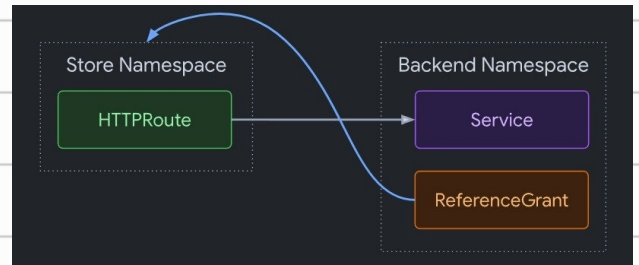
Eg: When HTTPRoute <sup>NS 'A'</sup> → <sup>NS 'B'</sup> Service

But why?

Team A: Website

Team B: Payment

Special whitelist  
access to "needed service only"



From Kubernetes Documentation

```
apiVersion: gateway.networking.k8s.io/v1
kind: ReferenceGrant
metadata:
  name: allow-ui-to-evolution
  namespace: pokedex-core
spec:
  from:
    - group: gateway.networking.k8s.io
      kind: HTTPRoute
      namespace: pokedex-ui
  to:
    - group: ""
      kind: Service
      name: evolution-engine
```

- This ReferenceGrant is named allow-ui-to-evolution in pokedex-core namespace

- We grant the reference permission to HTTPRoute in pokedex-ui namespace

- Only allow access to evolution-engine service

## Traffic mirroring

```
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: http-filter-mirror
  labels:
    gateway: mirror-gateway
spec:
  parentRefs:
    - name: mirror-gateway
  hostnames:
    - mirror.example
  rules:
    - backendRefs:
        - name: foo-v1
          port: 8080
      filters:
        - type: RequestMirror
          requestMirror:
            backendRef:
              name: foo-v2
              port: 8080
```

- Traffic mirroring

### Use when:

- Wants to test the system by production request but does not break / response to user

## Canary deployment [traffic splitting]

```
spec:
  rules:
    - backendRefs:
        - name: foo-v1
          port: 8080
          weight: 90
        - name: foo-v2
          port: 8080
          weight: 10
```

### Use when

- Ready for gradual rollout